



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1 2025/2026

SECJ3553 ARTIFICIAL INTELLIGENCE

SECTION 02

Assignment 2

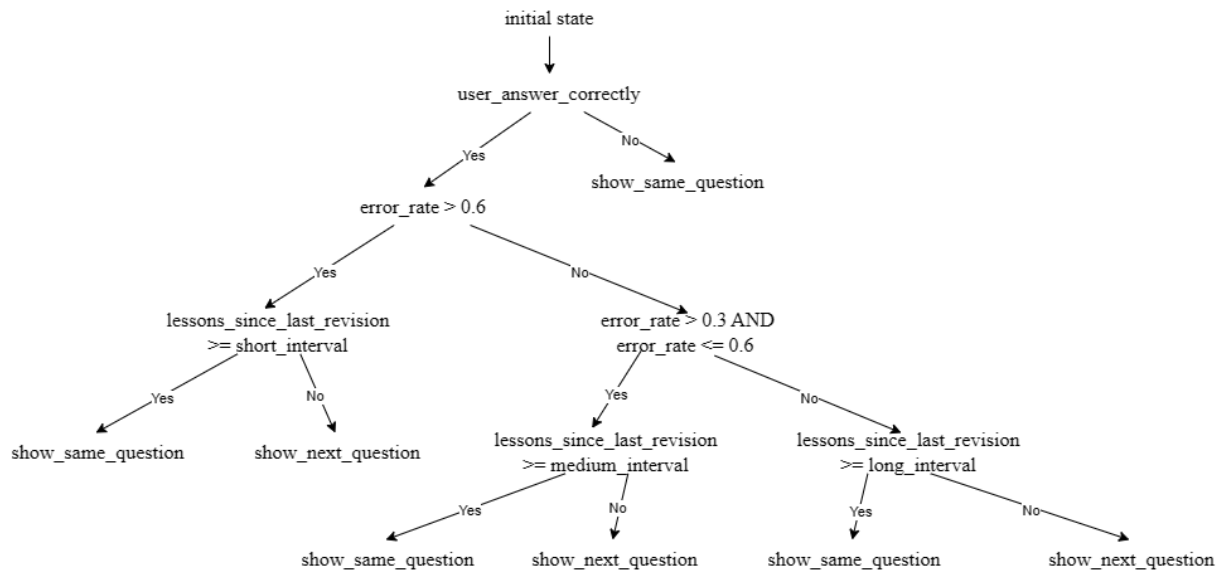
LECTURER: DR. ROLIANA BINTI IBRAHIM

GROUP

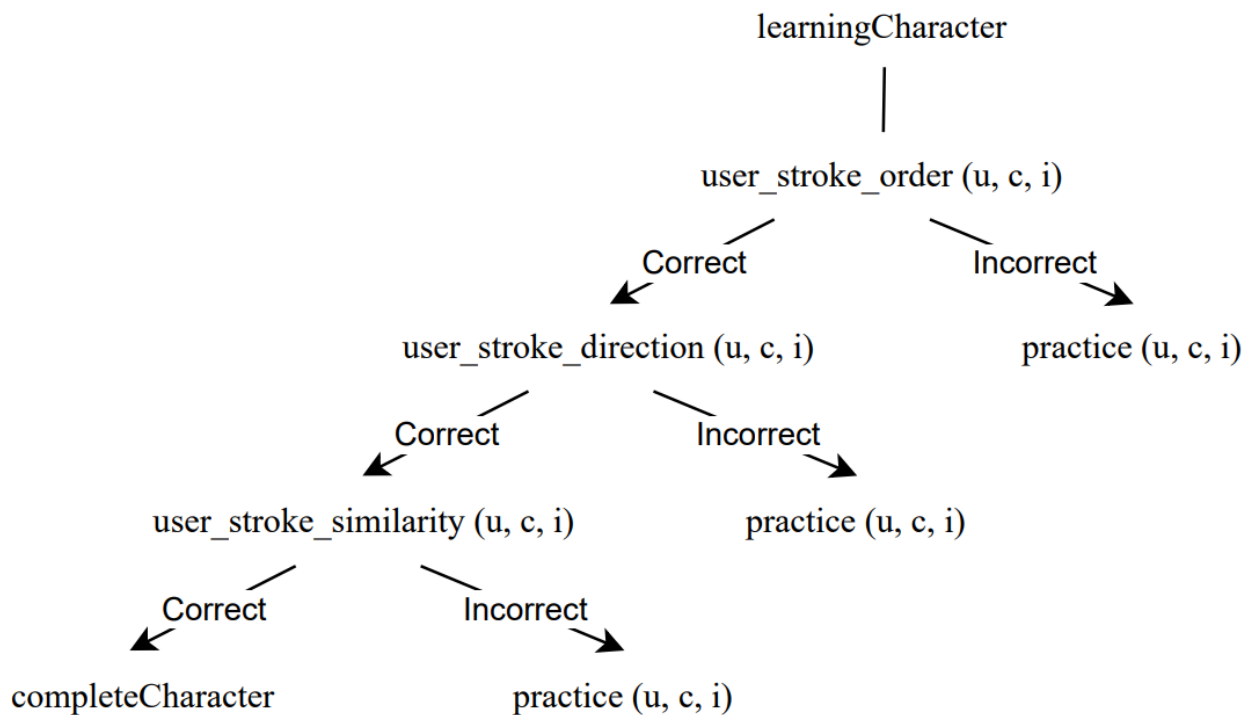
Student Name	Matric No.
DAMIYA AINA BINTI BASIR ABD SHAMMAD	A23CS0220
GOE JIE YING	A23CS0224
LAM YOKE YU	A23CS0233
TAN YI YA	A23CS0187
TEH RU QIAN	A23CS0191

1.0 State Space Search

1.1 Character Revision Frequency



1.2 Handwriting Recognition



States

learningCharacter

user_stroke_order (u, c, i)

user_stroke_direction (u, c, i)

user_stroke_similarity (u, c, i)

Actions

practice (u, c, i)

Goal

completeCharacter

Path Cost

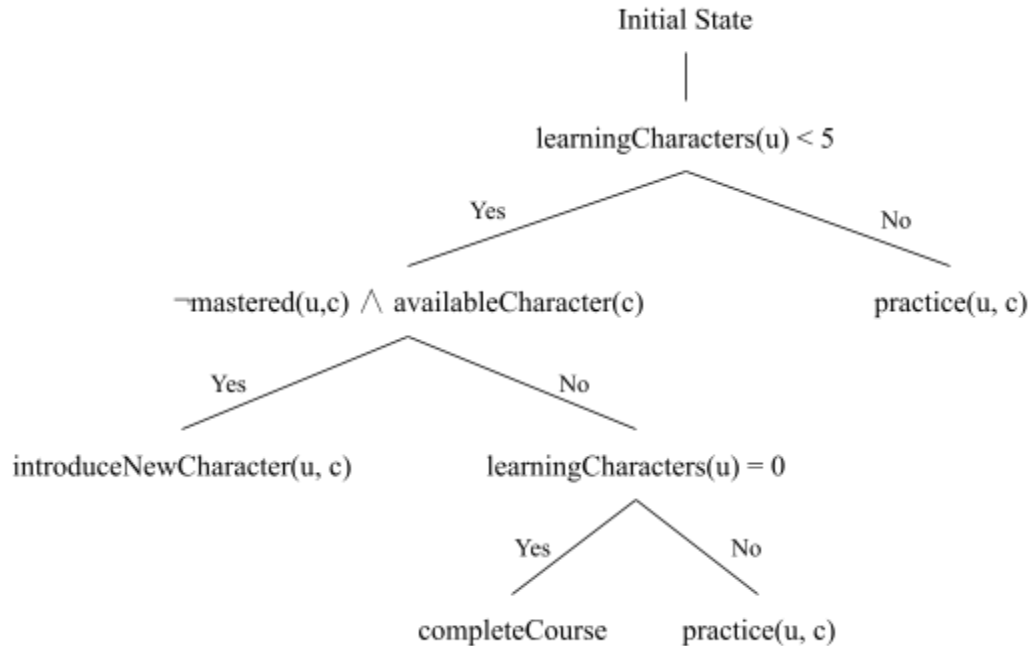
user_stroke_order (u, c, i) correct, user_stroke_direction (u, c, i) correct, user_stroke_similarity (u, c, i) correct

Solution

user_stroke_order (u, c, i) correct, user_stroke_direction (u, c, i) correct, user_stroke_similarity (u, c, i) correct

1.3 New Word Introduction

New characters shall be introduced from time to time to ensure progression. New characters are introduced when the user has demonstrated mastery of a current character by achieving a 90% accuracy rate or higher in their last 5 attempts. Each time there will be 5 characters alternating.



State

The state is the mastery of the user.

$$S = \{$$

$$\quad mastered(u, c): \text{boolean},$$

$$\quad accuracy(u, c): \text{int}[5],$$

$$\quad learningCharacters(u): \text{char}[]$$

$$\}$$

It may be impossible to plot all states given the varying characters and conditions, thus each ends that does not reach the goal would be directed to the root of the tree and repeat until the goal to complete the course is reached.

mastered(u, c)

This stores the characters c mastered by the user u . If a character has achieved a 90% accuracy rate of higher in their last 5 attempts, the character is considered mastered. The accuracy is computed in '1.2 Handwriting Recognition'.

accuracy(u, c)

This stores the accuracy score of the user u 's last 5 attempts on character c .

learningCharacters(u)

This function stores the list of current characters learned by the user u . These characters will be iterated based on the computation in ‘1.1 Character Revision Frequency’.

availableCharacter(c)

The system maintains a list of characters c that are available for learning and are sorted by their difficulty.

Actions

introduceNewCharacter(u, c)

If there exists a character c that has not been mastered ($\neg \text{Mastered}(u, c)$) and is available for learning (*AvailableCharacter(c)*), it will be selected as the next word to introduce.

practice(u, c)

The user will practice a character from the *learningCharacters(u)* list. The accuracy score would be updated. If the average accuracy score of the last 5 attempts is more than 90%, *mastered(u, c)* and *learningCharacters(u)* would be updated.

Goals

The goal is *completeCourse*, where all there are no characters that are not mastered and the learning characters list is empty.

Path Cost

Solution

1.4 Motivation & Reward Mechanism

1.5 Adaptive AI Feedback

2.0 How each of them work together